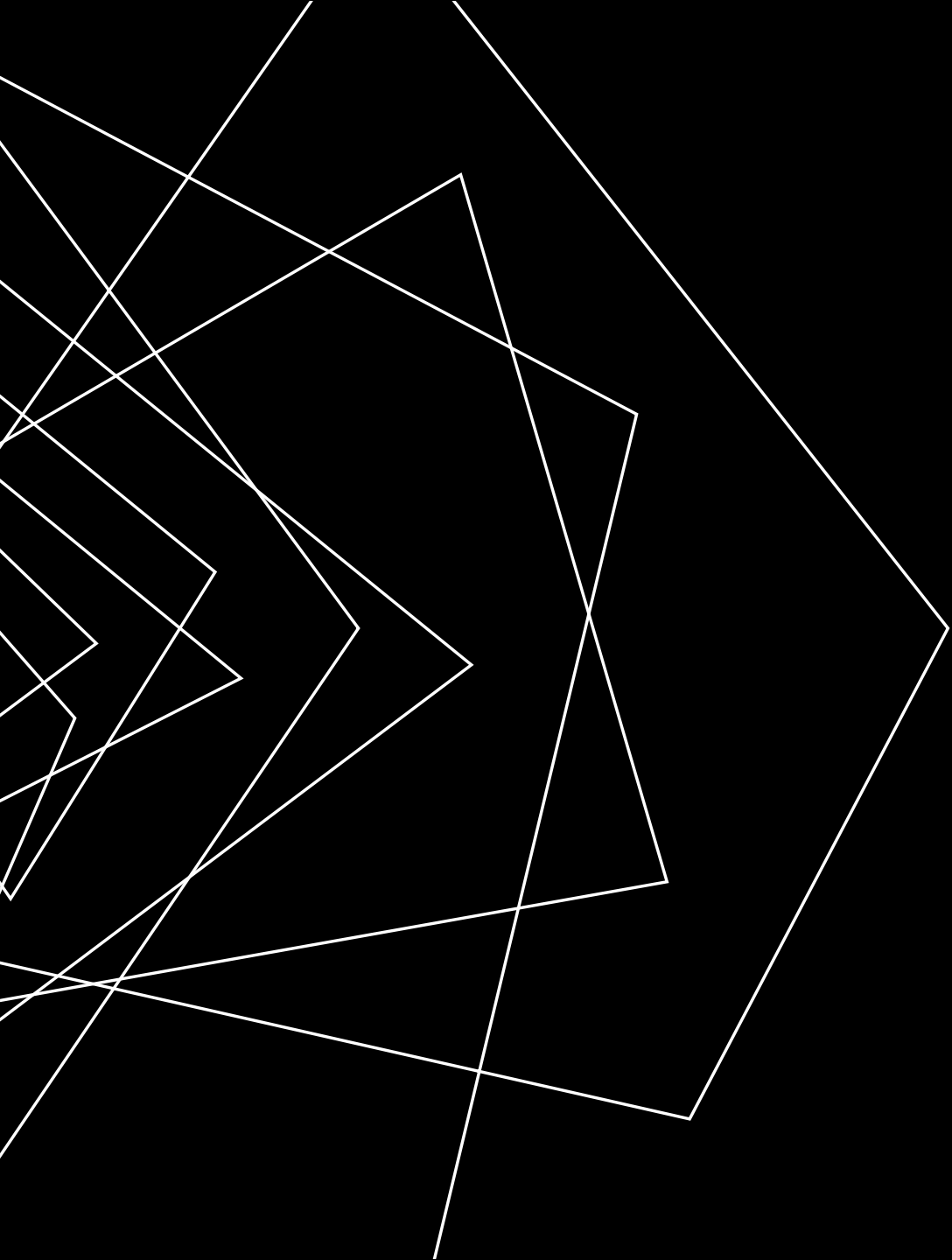




EAST TEXAS A&M
— UNIVERSITY —

CSCI 380.61E
WEB PROGRAMMING
AND INTERFACE DESIGN



REACTJS + VITE

Web Programming and Interface Design

LEARNING OBJECTIVES

- Understand what ReactJS is and why it's used
- Learn advantages of Single Page Applications (SPA)
- Explore Vite as a modern build tool
- Setup development environment with React + Vite

WHAT IS REACTJS?

- A JavaScript library for building user interfaces
- Developed and maintained by Facebook (Meta)
- Component-based architecture
- Virtual DOM for performance optimization

SPA VS TRADITIONAL DEVELOPMENT

- Traditional: Server reloads entire page
- SPA: Only updates the required components
- Faster, smoother user experience

WHY VITE?

- Lightning-fast cold starts
- Hot Module Replacement (HMR)
- Optimized build for production
- Simpler and faster than Create-React-App

SETUP ENVIRONMENT

- Install Node.js
- Install Vite: `npm create vite@latest my-app`
- Choose React template
- Run project: `npm run dev`

LAB EXERCISE

- Create a new React project using Vite
- Run the development server
- Edit App.jsx to display 'Hello World'
- Add a custom component and render it

LAB EXERCISE

- Create a simple React app with Vite
- Render 'Hello World' and a custom component
- Submit project files

LEARNING OBJECTIVES

- Understand React components
- Learn JSX syntax and usage
- Use props to pass data to components
- Build reusable UI components

REACT COMPONENTS

- Functional Components: Simple, use hooks
- Class Components: Older, less common
- Components are reusable building blocks

JSX BASICS

- Looks like HTML but is JavaScript syntax
- Can embed JavaScript expressions inside {}
- Example: `<h1>{title}</h1>`

PROPS IN REACT

- Props = Properties passed to components
- Allow reusability and customization
- Example: `<ProfileCard name='Alice' bio='Developer' />`

COMPONENT COMPOSITION

- Nest components inside each other
- Helps build complex UIs from small pieces
- Example: StudentList contains multiple StudentCard components

CODE EXAMPLE - PROFILECARD

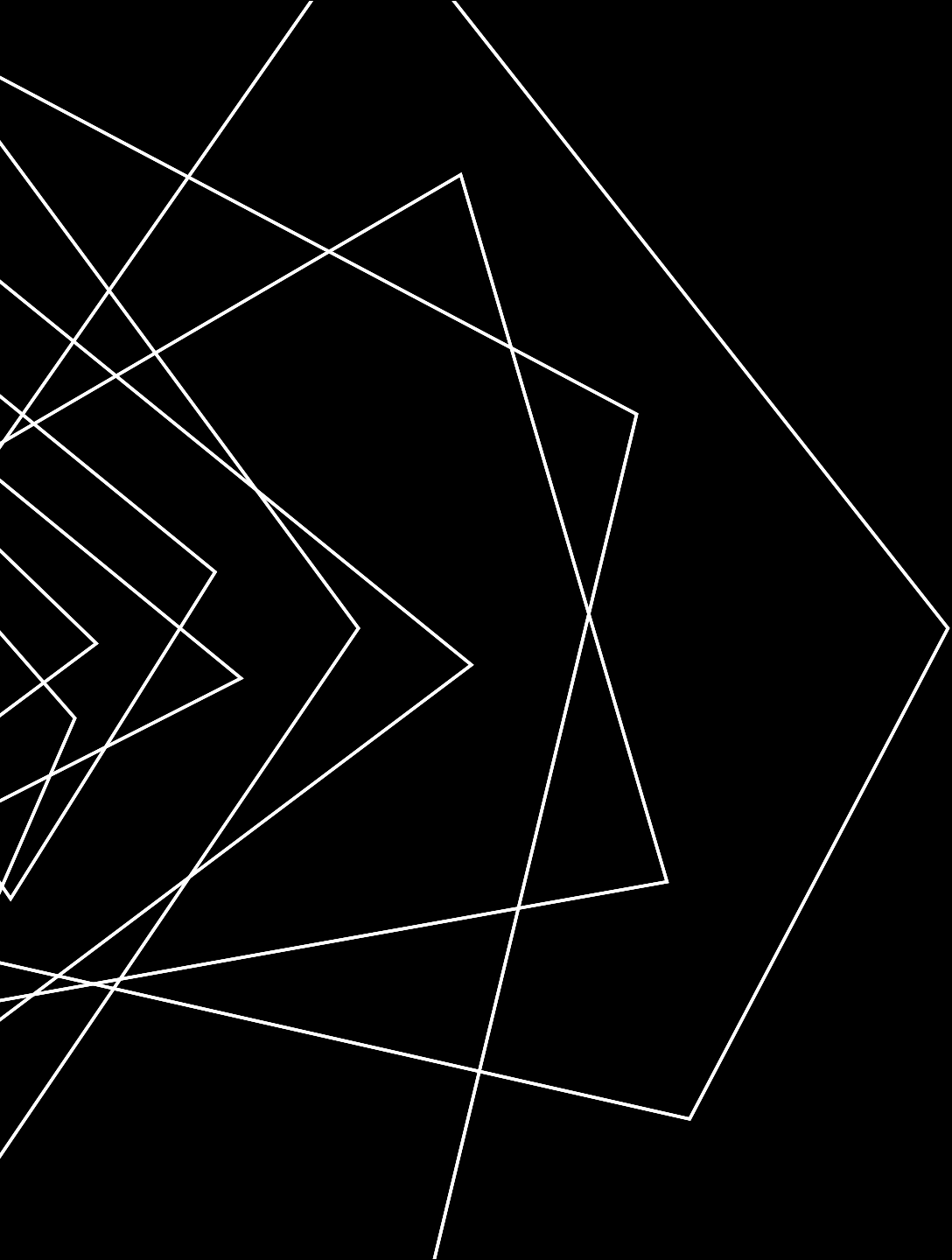
```
function ProfileCard(props) {  
  return (  
    <div>  
      <h2>{props.name}</h2>  
      <p>{props.bio}</p>  
    </div>  
  );  
}
```

LAB EXERCISE

- Build a ProfileCard component with props: name, bio, image
- Render multiple ProfileCards in App.jsx
- Experiment with different prop values

LAB EXERCISE

- Create a StudentList component
- Render multiple student profiles using props



THANK YOU

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